

Also, $\psi^{-\dagger}$ and ψ^+ would annihilate a vacuum state on the right, and ψ^- and $\psi^{+\dagger}$ would annihilate the vacuum state on the left, so ψ^+ and ψ^- may be replaced everywhere in Eq. (6.2.29) with the complete field $\psi = \psi^+ + \psi^-$:

$$-i\Delta_{\ell m}(x, y) = \theta(x - y)\langle\psi_{\ell}(x)\psi_m^{\dagger}(y)\rangle_0 \pm \theta(y - x)\langle\psi_m^{\dagger}(y)\psi_{\ell}(x)\rangle_0. \quad (6.2.30)$$

This is often written

$$-i\Delta_{\ell m}(x, y) = \langle T\{\psi_{\ell}(x)\psi_m^{\dagger}(y)\}\rangle_0, \quad (6.2.31)$$

where T is a time-ordered product, whose definition is now extended** to all fields, with a minus sign for any odd permutation of fermionic operators.

6.3 Momentum Space Rules

The Feynman rules outlined in Section 6.1 specify how to calculate the contribution to the S -matrix of a given N th order diagram, as the integral over N spacetime coordinates of a product of spacetime-dependent factors. For a final particle (or antiparticle) line with momentum p'^{μ} leaving a vertex with spacetime coordinate x^{μ} , we get a factor proportional to $\exp(-ip' \cdot x)$, and for an initial particle line with momentum p^{μ} entering a vertex with spacetime coordinate x^{μ} , we get a factor proportional to $\exp(+ip \cdot x)$. In Section 6.2 we saw that the factor associated with an internal line running from y to x can be expressed as a Fourier integral, over off-shell four-momenta q^{μ} , of an integrand proportional to $\exp(iq \cdot (x - y))$. We can think of q^{μ} as the four-momentum flowing along the internal line in the direction of the arrow from y to x . Hence the integral over each vertex's spacetime position merely yields a factor

$$(2\pi)^4 \delta^4 \left(\sum p + \sum q - \sum p' - \sum q' \right), \quad (6.3.1)$$

where $\sum p'$ and $\sum p$ denote the total four-momentum of all the final or initial particles leaving or entering the vertex, and $\sum q'$ and $\sum q$ denote the total four-momentum of all the internal lines with arrows leaving or entering the vertex, respectively. Of course, in place of these integrals over x^{μ} s, we now have to do integrals over the Fourier variables q^{μ} , one for each internal line.

These considerations can be encapsulated in a new set of Feynman rules

** This is not inconsistent with our previous definition of the time-ordered product of Hamiltonian densities in Chapter 3, because the Hamiltonian density can only contain even numbers of fermionic field factors.